

Sample End Semester Question Paper - 2025

Subject: Machine Learning | Code: CS601 | Semester: 6

Branch: CSE | Regulation: R20 (sample) | Time: 3 Hours | Max Marks:

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. State Bayes' theorem.
2. State one advantage of k-NN.
3. What is a kernel in SVM?
4. What is backpropagation?
5. What is a confusion matrix?
6. Define machine learning.
7. State one example of supervised learning.
8. What is overfitting?
9. State the decision rule of logistic regression.
10. What is information gain?

PART B - Long questions (8 x 10 = 80 Marks)

1. Explain k-NN classification and K-means clustering with examples.
2. Explain SVM and the kernel trick.
3. Explain neural networks: perceptron and backpropagation with an example.
4. Explain model evaluation: metrics, cross-validation and confusion matrix.
5. Explain linear and logistic regression with derivation and an example.
6. Explain the bias-variance trade-off and overfitting with remedies.
7. Explain decision trees (ID3/CART) with an example.
8. Explain Naive Bayes classification with a worked example.